Lathika N 241801135 exp 2

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv('/content/WA_Fn-UseC_-Telco-Customer-Churn.csv')

df['TotalCharges'] = pd.to_numeric(df['TotalCharges'], errors='coerce')
df['TotalCharges'] = df['TotalCharges'].fillna(df['TotalCharges'].median())
df = df.drop_duplicates()

print("Shape:", df.shape)
print("\nMissing Values:\n", df.isnull().sum())
print("\nFirst 5 Rows:")
display(df.head())
```

```
Shape: (7043, 21)

Missing Values:
 customerID          0
gender              0
SeniorCitizen       0
Partner             0
Dependents          0
tenure              0
PhoneService        0
MultipleLines       0
InternetService     0
OnlineSecurity      0
OnlineBackup        0
DeviceProtection    0
TechSupport         0
StreamingTV         0
StreamingMovies     0
Contract            0
PaperlessBilling    0
PaymentMethod       0
MonthlyCharges      0
TotalCharges        0
Churn               0
dtype: int64

First 5 Rows:
```

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecur
0	7590-VHVEG	Female	0	Yes	No	1	No	No phone service	DSL	
1	5575-GNVDE	Male	0	No	No	34	Yes	No	DSL	
2	3668-QPYBK	Male	0	No	No	2	Yes	No	DSL	
3	7795-CFOCW	Male	0	No	No	45	No	No phone service	DSL	
4	9237-HQITU	Female	0	No	No	2	Yes	No	Fiber optic	

5 rows × 21 columns

```
fig, ax = plt.subplots(2, 2, figsize=(14, 9))

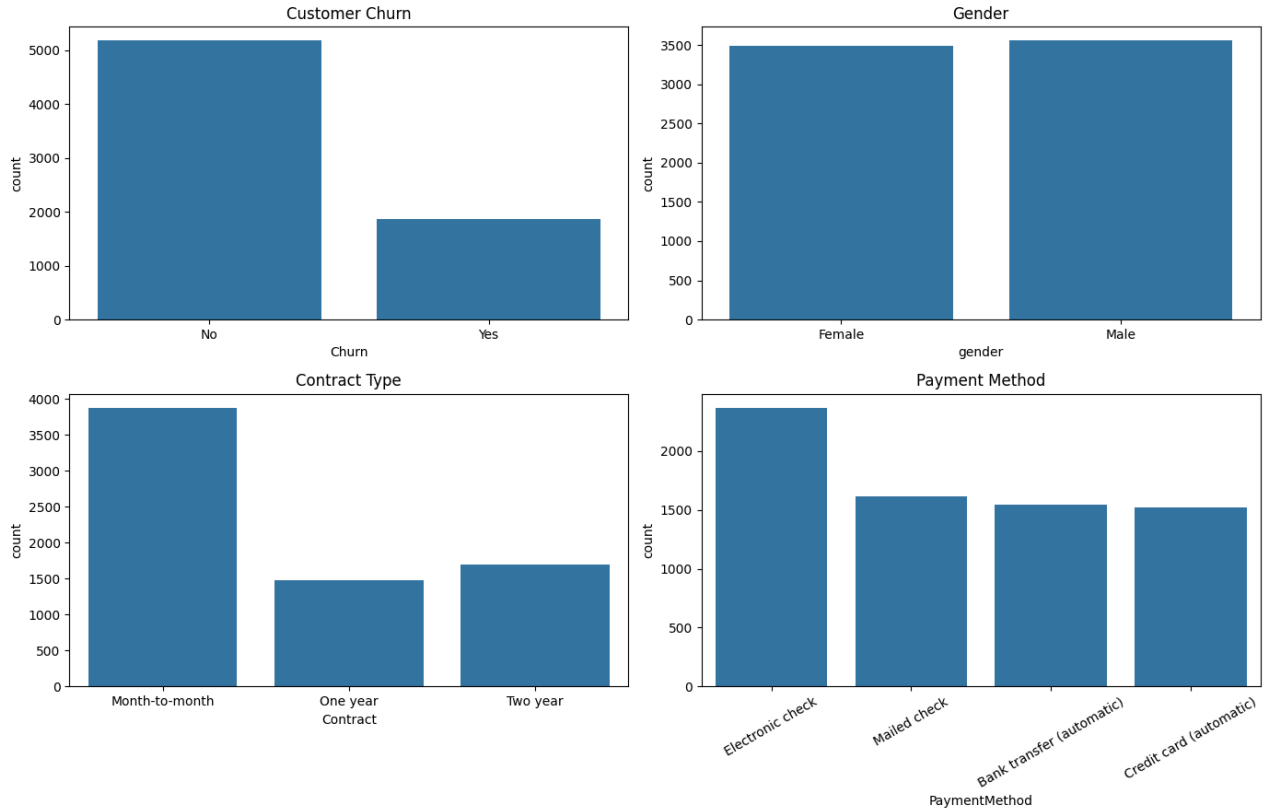
sns.countplot(x='Churn', data=df, ax=ax[0,0])
ax[0,0].set_title('Customer Churn')

sns.countplot(x='gender', data=df, ax=ax[0,1])
ax[0,1].set_title('Gender')

sns.countplot(x='Contract', data=df, ax=ax[1,0])
ax[1,0].set_title('Contract Type')

sns.countplot(x='PaymentMethod', data=df, ax=ax[1,1])
ax[1,1].set_title('Payment Method')
ax[1,1].tick_params(axis='x', rotation=30)

plt.tight_layout()
plt.show()
```

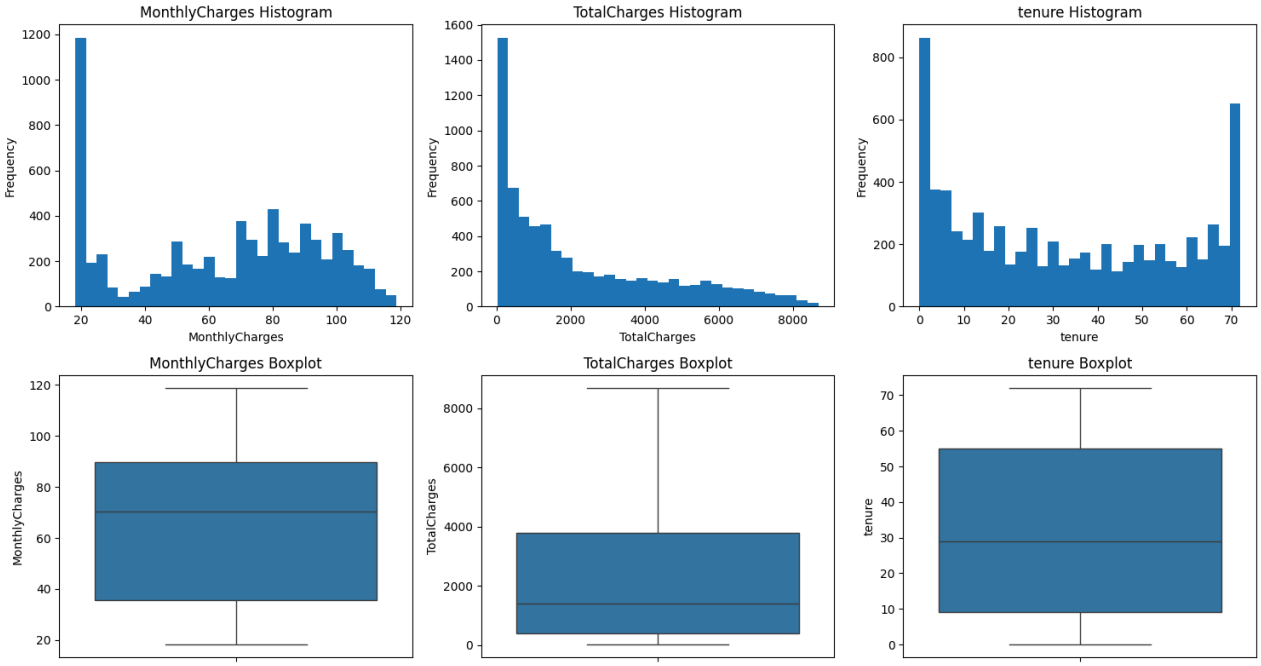



```
fig, ax = plt.subplots(2, 3, figsize=(15, 8))

for i, col in enumerate(['MonthlyCharges', 'TotalCharges', 'tenure']):
    ax[0,i].hist(df[col], bins=30)
    ax[0,i].set_title(col + ' Histogram')
    ax[0,i].set_xlabel(col)
    ax[0,i].set_ylabel('Frequency')

    sns.boxplot(y=df[col], ax=ax[1,i])
    ax[1,i].set_title(col + ' Boxplot')

plt.tight_layout()
plt.show()
```




```
for col in ['MonthlyCharges', 'TotalCharges', 'tenure']:
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
    IQR = Q3 - Q1

    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR

    outliers = df[(df[col] < lower) | (df[col] > upper)]

    print("\n", col)
    print("Q1 =", Q1)
    print("Q3 =", Q3)
    print("IQR =", IQR)
    print("Lower Bound =", lower)
    print("Upper Bound =", upper)
    print("Number of Outliers =", len(outliers))
```

```
 MonthlyCharges
Q1 = 35.5
Q3 = 89.85
IQR = 54.349999999999994
Lower Bound = -46.02499999999999
Upper Bound = 171.375
Number of Outliers = 0

 TotalCharges
Q1 = 402.225
Q3 = 3786.6
IQR = 3384.375
Lower Bound = -4674.3375
Upper Bound = 8863.1625
Number of Outliers = 0

 tenure
Q1 = 9.0
Q3 = 55.0
IQR = 46.0
Lower Bound = -60.0
Upper Bound = 124.0
Number of Outliers = 0
```

```
fig, ax = plt.subplots(2, 2, figsize=(14, 10))

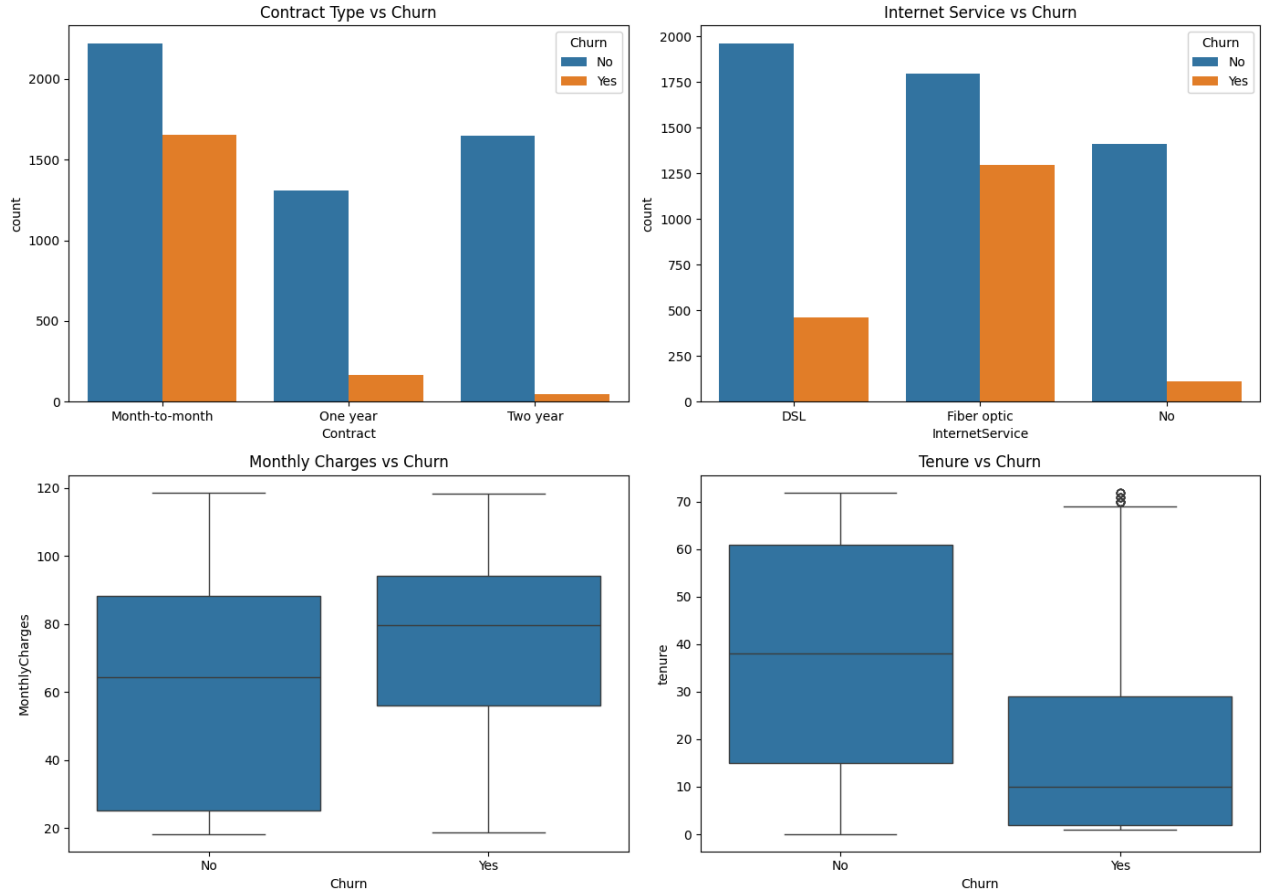
sns.countplot(x='Contract', hue='Churn', data=df, ax=ax[0,0])
ax[0,0].set_title('Contract Type vs Churn')

sns.countplot(x='InternetService', hue='Churn', data=df, ax=ax[0,1])
ax[0,1].set_title('Internet Service vs Churn')

sns.boxplot(x='Churn', y='MonthlyCharges', data=df, ax=ax[1,0])
ax[1,0].set_title('Monthly Charges vs Churn')

sns.boxplot(x='Churn', y='tenure', data=df, ax=ax[1,1])
ax[1,1].set_title('Tenure vs Churn')

plt.tight_layout()
plt.show()
```




```
print("1. CONTRACT TYPE WITH HIGHEST CHURN")
print(pd.crosstab(df['Contract'], df['Churn'], normalize='index')['Yes'].sort_values(ascending=False))

print("\n2. PAYMENT METHOD MOST USED BY CHURNED CUSTOMERS")
print(df[df['Churn']=='Yes']['PaymentMethod'].value_counts())

print("\n3. AVERAGE TENURE BY CHURN")
print(df.groupby('Churn')['tenure'].mean())

print("\n4. AVERAGE MONTHLY CHARGES BY CONTRACT")
print(df.groupby('Contract')['MonthlyCharges'].mean())

print("\n5. INTERNET SERVICE WITH HIGHEST CHURN")
```

```
print(pd.crosstab(df['InternetService'], df['Churn'], normalize='index')['Yes'].sort_values(ascending=False))

print("\n6. MONTHLY CHARGES BY CHURN")
print(df.groupby('Churn')['MonthlyCharges'].mean())

print("\n7. IMPORTANT CHURN INDICATORS")
```

```
1. CONTRACT TYPE WITH HIGHEST CHURN
Contract
Month-to-month    0.427097
One year          0.112695
Two year          0.028319
Name: Yes, dtype: float64

2. PAYMENT METHOD MOST USED BY CHURNED CUSTOMERS
PaymentMethod
Electronic check             1071
Mailed check                  308
Bank transfer (automatic)     258
Credit card (automatic)       232
Name: count, dtype: int64

3. AVERAGE TENURE BY CHURN
Churn
No     37.569965
Yes    17.979133
Name: tenure, dtype: float64

4. AVERAGE MONTHLY CHARGES BY CONTRACT
Contract
Month-to-month    66.398490
One year          65.048608
Two year          60.770413
Name: MonthlyCharges, dtype: float64

5. INTERNET SERVICE WITH HIGHEST CHURN
InternetService
Fiber optic    0.418928
DSL            0.189591
No             0.074050
Name: Yes, dtype: float64

6. MONTHLY CHARGES BY CHURN
Churn
No     61.265124
Yes    74.441332
Name: MonthlyCharges, dtype: float64

7. IMPORTANT CHURN INDICATORS
Contract, Internet Service, Tenure, Monthly Charges and Payment Method
```

```
from sklearn.model_selection import train_test_split

df_encoded = pd.get_dummies(df, drop_first=True)

X = df_encoded.drop('Churn_Yes', axis=1)
y = df_encoded['Churn_Yes']

X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.20,
    random_state=42,
    stratify=y
)

df_encoded.to_csv('/content/CustomerChurn_Preprocessed.csv', index=False)

print("Training Data:", X_train.shape)
print("Testing Data:", X_test.shape)
print("\nCustomerChurn_Preprocessed.csv saved successfully!")
```

```
Training Data: (5634, 7072)
Testing Data: (1409, 7072)
```